

Sandisk ATMOS AI Appliance Portfolio

Sell a complete private-AI service, not an E3.S accelerator kit. Leadership product and proof proposal.

CERTIFIED APPLIANCE

4 / 8 / 16 / 24-device family
OEM, topology, thermals and serviceability

TURNKEY SOFTWARE

AI OS + Model Services + Operator
Serving Manager, SDK and Control Center

KNOWLEDGE AGENT

Ingest -> retrieve -> rerank -> cite
Governed tools, evaluation and audit

ONE SUPPORT ROUTE

Signed packs, upgrades and rollback
Sandisk triage + coordinated OEM support

REQUEST: approve 8/16-device POCs, flagship Knowledge Agent, software ownership and parallel OEM feasibility.

Named SKUs. Explicit system and service profiles.

Five commercial profiles share four device densities; Enterprise 8 adds production integration and serviceability.

Configuration	Devices / groups	HBF nominal / study	DRAM / NVMe	Service network
Edge Workstation 4	4 / 1	512 / 400 GB	128-256 GB / 2-4 TB	25/100 GbE
Departmental 8	8 / 2	1.024 / 0.800 TB	256-512 GB / 4-8 TB	100 GbE
Enterprise 8	8 / 2	Same device envelope	Qualified 8-device BOM	Enterprise 8 BOM
Enterprise 16	16 / 4	2.048 / 1.600 TB	512 GB-1 TB / 8-16 TB	2 x 100/200 GbE
Enterprise 24 / stretch	24 / 6	3.072 / 2.400 TB	1 TB+ / 16 TB+	200/400 GbE

PROFILE DIFFERENCES / CUSTOMER RECEIVES

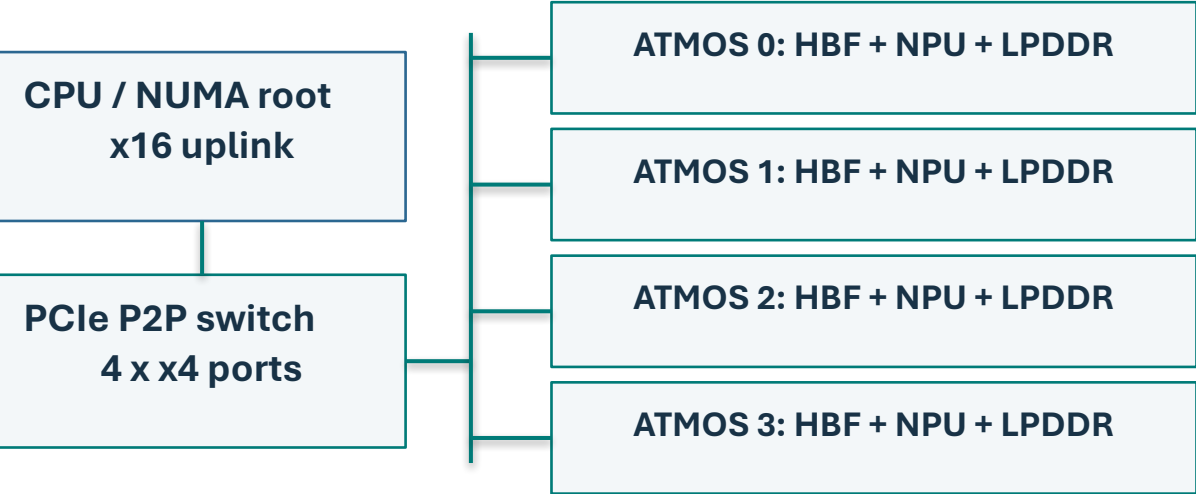
Workstation: local catalog + APIs. Departmental: projects, quotas and separate queues. Enterprise: identity, audit, backup and support.

CPU, form factor, redundant power and acoustic/air/DLC qualification are OEM-specific. Aggregate HBF is distributed, not coherent. PRD 2 / 15.

Four-device domains: local work before peer exchange

One CPU/NUMA root -> x16 uplink -> qualified switch -> four x4 ATMOS modules. Group placement is a software contract.

STANDARD SWITCH-LOCAL ALLOCATION DOMAIN



PER-DEVICE PLANNING ENVELOPE

Resource	Target / constraint
HBF	128 GB nominal / 100 GB study
Local read / write	~200 GB/s / ~64 TOPS-class
Dynamic state	8-16 GB LPDDR + SRAM
Module / link	E3.S 2T / Gen5 or Gen6 x4
Sustained power	30-40 W; accelerator duty

4 devices / 0.512 TB Device-only power: 120-160 W	8 devices / 1.024 TB Device-only power: 240-320 W	16 devices / 2.048 TB Device-only power: 480-640 W	24 devices / 3.072 TB Device-only power: 720-960 W
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P2P is not coherent memory or local HBF bandwidth. Qualify Gen5/Gen6 separately, per-device fit, IOMMU, safe reset and cross-domain traffic. PRD 2 / 12.

The software is the product: deliver an integrated stack

Packaged components, stable contracts and operations—not a requirement for the customer to assemble an inference platform.

EXPERIENCE AND KNOWLEDGE

Knowledge Agent / UI
Search, RAG, approved workflows

SDK / API portal / notebooks
Python, REST, examples and profiling

Knowledge plane
Connectors, parsers, ACLs and indexes

Certified ModelPacks
Weights, tokenizer, kernels and tests

SERVING AND DEVICE EXECUTION

Gateway
Auth, quota, API policy and accounting

Serving Manager
KServe/llm-d candidate; replicas, queues

Model Services / ATMOS runtime
Certified operators and memory plans

AI OS + Device Service
DRA/CDI/operator, DMA, P2P, firmware

TRUST AND FLEET OPERATIONS

Control Center
Fleet, SLO, updates and cases

Identity / secrets / network
OIDC/SAML, RBAC, vault, TLS/mTLS

Supply chain / policy / evidence
Signing, SBOM, policy and audit

Enterprise Support
Prod/LTS, CVEs, OEM escalation

ATMOS runtime/driver/operator, HBF placement and workload adapters must be delivered/qualified; upstream projects do not establish ready ATMOS support. PRD 3-4 / 12.

Seven service APIs. A bounded certified workload catalog.

Support follows model, input, candidate, policy and SLO certification—not compatibility by endpoint name alone.

API family	Proposed surface / service contract
OpenAI-compatible	/v1/chat/completions, /responses, /models; certified params
Embeddings	/v1/embeddings; batch/indexing; model and input limits
Retrieval	/v1/retrieve; hybrid search, collections, filters and metadata
Reranking	/v1/rerank; batch; certified candidate limits
Documents	Ingest, status, delete, reindex, version and extraction results
Agents	Sessions/runs, approved tools, citations and policy output
Administration	Projects/users/keys/quotas, models, updates and support

CERTIFY FIRST / LAUNCH CANDIDATES

Embeddings, hybrid retrieval, reranking, cited Knowledge Agent and qualified document services.

MODEL-SPECIFIC PILOTS / RESEARCH

7B-32B private LLM: quality + TTFT/TPOT/context proof.
Recommendation features: pilot. MoE prefill/cold experts: research/pilot.

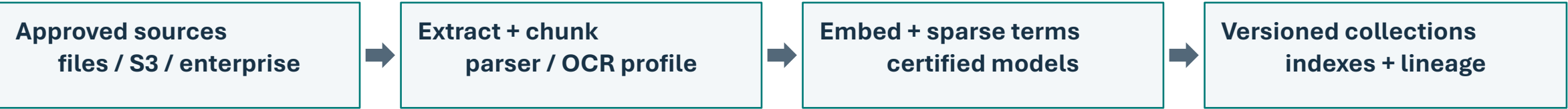
Not launch targets: long-context dense decode, training, graphics or safety-critical control.

GA candidate is a target, not a released certification. Healthcare/industrial packs provide knowledge assistance, not diagnostic or control claims. PRD 4 / 16.

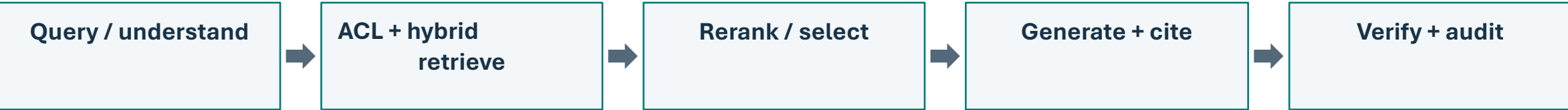
Out-of-box Knowledge Agent: governed evidence to answers

The flagship solution ships ingestion, hybrid retrieval, reranking, citations, evaluation and bounded agent workflows.

INGESTION / SOURCE, VERSION, TENANT AND ACL LINEAGE



ONLINE QUERY / ACL FILTERING BEFORE EVIDENCE REACHES GENERATION



SIX INCLUDED MODES / SELECT WITHIN THE CERTIFIED ENVELOPE

Search	Ranked passages only	Standard RAG	One retrieval / answer pass	Deep RAG	Decompose, broaden, synthesize
Agentic RAG	Plan, tools, retry, verify	Retrieval-only	Evidence to an existing LLM	Batch knowledge	Summaries, extract, tag

Agentic mode adds model calls, latency and budget risk. Search/retrieval-only can return evidence without ATMOS generation. Full generation requires model certification. PRD 5-8.

Governed by default. Configurable within certified limits.

Tenant/source ACLs must filter retrieval before generation; post-generation masking is not the security boundary.



CUSTOMER CONFIGURATION

Collections, source schedules and retention. Prompts/citations and certified retrieval settings. Approved tools, models, modes, quotas and golden tests.

SUPPORTED EXTENSIONS

REST/OpenAI + Python SDK, webhooks and approved MCP. Signed restricted connectors and supported workflows. BYO models through Sandisk certification.

LOCKED PRODUCTION

Firmware, driver, kernels and switch policy. Unsigned binaries, hidden tokenizer/precision changes, unbounded tools and cross-tenant P2P are not self-service.

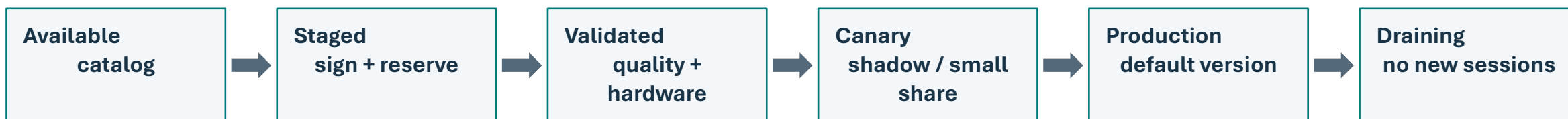
BASE TOOLS: read-only search/fetch/lookup; sandboxed calculator; draft-only messaging. Actions need approval; code execution is not in base.

Trust: signed firmware/images/ModelPacks, SBOM/CVE, TPM/secure boot, vault, TLS/mTLS, retention/deletion and audit. No unearned compliance certification. PRD 6-8 / 12.

Model and platform upgrades are controlled product releases

ModelPack binds weights, tokenizer, preprocessing, precision, compiler output, runtime, golden tests, APIs and SLO.

MODELPACK STATE MACHINE



Failed canary -> Rolled back. Draining -> Retired. Active sessions remain generation-pinned; preserve a rollback generation.

GENERATOR / RERANKER UPDATE

Stage -> compatibility/golden tests -> shadow/canary -> compare quality/p99/errors -> promote gradually; retain rollback.

EMBEDDING OR INDEX-COMPATIBILITY CHANGE

Build/re-embed a shadow collection -> retrieval evaluation / optional dual-read -> atomic generation switch -> hold old index.

PLATFORM UPDATE / ADVANCE ONE FAILURE DOMAIN AT A TIME



Offline: signed bundles, customer windows, no cloud required. Breaking parser/index/model/policy changes need migration + rollback. PRD 9-10.

Day-1 setup. Day-2 operation. One support route.

First-service setup within one business day is a target; corpus ingestion time depends on data volume, connectors and policy.



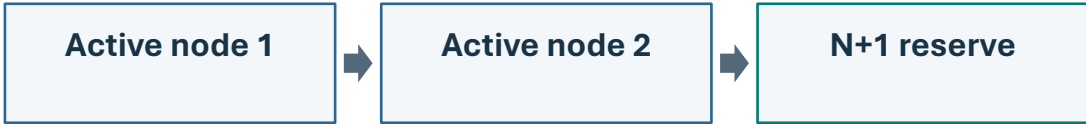
CONTROL CENTER / CORRELATED FLEET AND SERVICE VIEWS

Fleet BOM, serial, warranty, entitlement	Devices HBF/NPU, health, resets, thermal
Topology NUMA, PCIe width, switch/P2P	Services Models, sessions, queues, SLO
Knowledge Ingestion, collections, generations	Tenants Usage, quota, keys, chargeback
Updates Canary, drift, compatibility, rollback	Support Alerts, evidence, case, OEM dispatch

Release targets: **Production 3y | LTS 5y | optional +2y.**
Vertical 7-10y only with a funded business case; not contracted.

Support/lifecycle durations are proposed targets, not contracted entitlements. Evidence bundles redact inventory, versions, SLO/error, thermal and audit data. PRD 3 / 10-11.

HA REQUIRES COMPLETE REPLICAS AND PEAK RESERVE



One node: no service HA. Two nodes: replicas, not guaranteed full peak after a loss. Three-node N+1: two-node active capacity plus reserve.

SUPPORT OWNERSHIP

Sandisk: first call + OS/device/runtime/packs. OEM: chassis/platform, coordinated entitlement. NPU vendor and certified ISV: named escalation.

Adapt an established OEM platform—qualify the whole BOM

Parallel feasibility, not a partner commitment: early engineering flexibility, enterprise service and regulated lifecycle.

Candidate	PRD rationale / public platform	Product role	ATMOS qualification risk
Supermicro	X14 Petascale / E3.S-rich 24-32 class	Early 8/16 engineering; flexible configuration	2T/x4/bay power, P2P backplane and productized support
Dell	PowerEdge R7725 / R7725xd; dense E3.S and root maps	Enterprise channel, OpenManage and field service	Direct CPU attachment may need a new switch/P2P backplane
HPE	DL380/DL345 Gen12; iLO, cooling and x4 cabling options	Regulated edge, lifecycle and management	Dedicated thermal BOM; GPUs/NICs/DIMMs constrain ambient

QUALIFICATION LADDER / DO NOT INFER SUPPORT FROM STORAGE BAY COUNT



Sign exact 2T/x4 topology, sustained bay/switch cooling, BIOS/BMC, DMA isolation, safe replacement, spares, NRE and prime support. PRD 13-15.

Counterpart positioning and the conditional cost target

The only NVIDIA comparison page: workload-specific positioning, not a universal replacement or a measured price/performance claim.

PRD reference	Published reference memory	Closest SKU / limit
DGX Spark	128 GB unified; 273 GB/s	W4 / A8; capacity is not compute parity
RTX PRO 6000	96 GB GDDR7	A8; no graphics/training parity
DGX Station	748 GB coherent; HBM up to 7.1 TB/s	A8; much lower local compute/bandwidth
RTX PRO Server	2/4/8 x 96 GB; ~1.6 TB/s per GPU	E8/16/24; certify the exact service
IGX Thor	128 GB + optional dGPU	E8/16 backend, not a safety/controller replacement

70% ACQUISITION: DESIGN-TO-COST OBJECTIVE

24.8%

Illustrative 3-year TCO saving
1.218N versus 1.62N

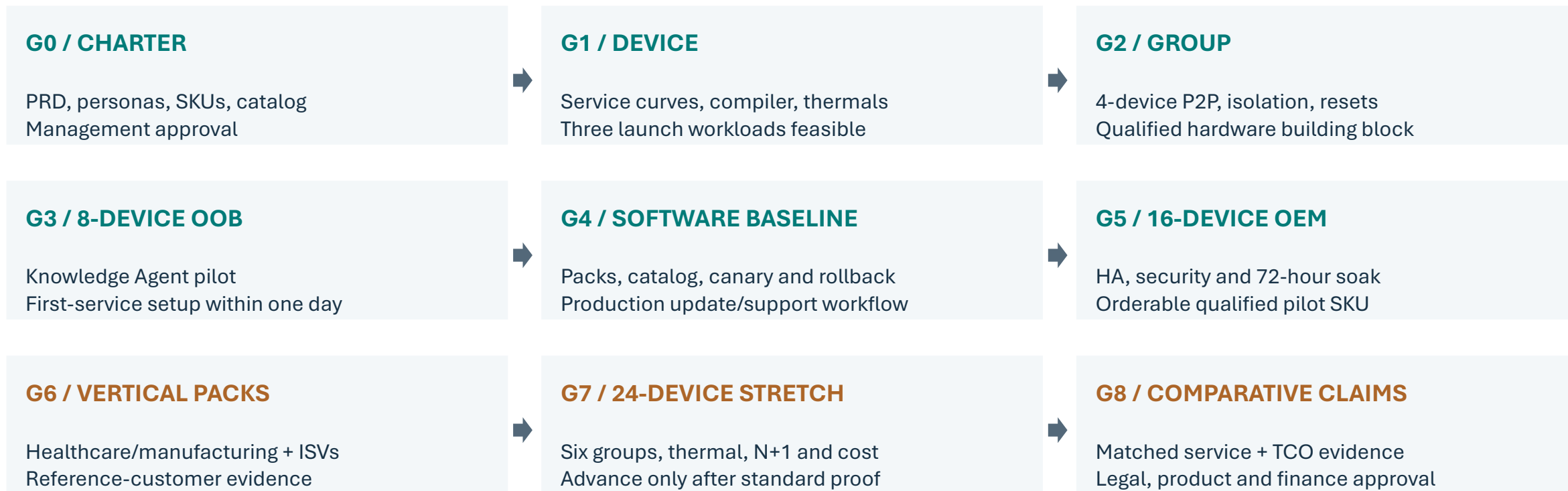
Synthetic 4-GPU-server-class N=\$180k:
\$126k ATMOS acquisition; 3-year TCO
\$219,240 versus \$291,600.

Break-even: <1.33 ATMOS systems per counterpart. At 1.25x: ~6% saving. At 2x: ~50% higher TCO.

PRD scenario only: same quality, SLO, peak demand and availability; 3y undiscounted. All prices synthetic; use quotes, measured node counts and removable spend. PRD 17-18.

Release gates convert a PRD into a supportable product

Evidence sequence, not a calendar promise. Approve claims only after certified quality, service, availability and economics.



First POCs: 8-device Knowledge Agent / 1-5m pages or chunks; 16-device retrieval/rerank; 8-device private AI; thermal and lifecycle tests.

Fail on quality/ACL/safety, unsafe DMA, missed p99/freshness or peak HA, thermal derating, incomplete software/support, or adverse accepted-service TCO. PRD 19-20.

Approve bounded product proof—not a launch claim

Leadership needs scope, accountable software/service owners, an OEM qualification path and a finance gate.

APPROVE / ASSIGN / HOLD

PRODUCT	Approve 8/16-device POCs and the OOB Knowledge Agent; retain strict workload and claims boundaries.
SOFTWARE	Name owners for AI OS, driver/runtime, Model Services/Operator, Control Center, SDK and release/support.
PLATFORM	Start Supermicro/Dell/HPE feasibility; freeze 2T/x4/P2P, thermal, HA, firmware and service contracts.
COMMERCIAL	Keep 70% price as a target; require quotes, certification staffing, measured unit costs and cashable displacement.
HOLD	24-device launch, unsupported models, compliance/diagnostic/safety claims and universal replacement promises.

CLICK TO OPEN PRD DETAILS

- 2 / 15 Hardware and SKU specs
- 3-4 Turnkey stack and APIs
- 5-8 Knowledge, data, tools, safety
- 9-10 Models and platform upgrades
- 11-12 Operations, support, security
- 13-15 OEM design and qualification
- 16 Workload and vertical catalog
- 17-18 Counterparts and TCO
- 19-20 Gates, risks and decisions
- A PRD public source register

Keep PPTX + linked PRD together. Each slide offers a full-document link if the viewer ignores section bookmarks. Original PRD content is preserved.